Rational - Canonical Form
Jordan - Canonical Form
Smith - Canonical Form.

Positive form
bilinear form
Matrix Analysis. (Matrix Norm)
Determinant.

Fourier series
Integral and differential form
More topics in series and sequence of functions.
Differential equation.

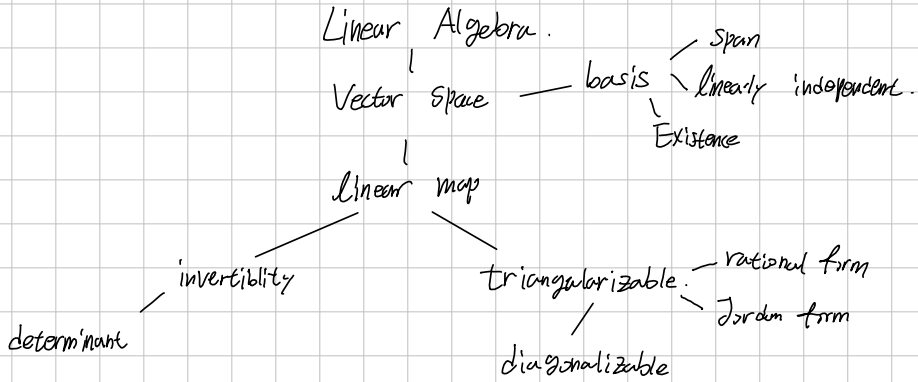
Goal

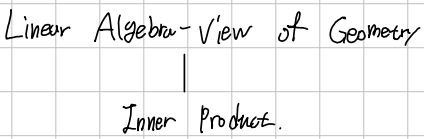
1. All Problem in:
   Dummit,
   Hoffman,
   Rudin, KKG
   Munkres,
   Hutcher,
   O'neil,
   Silverman.

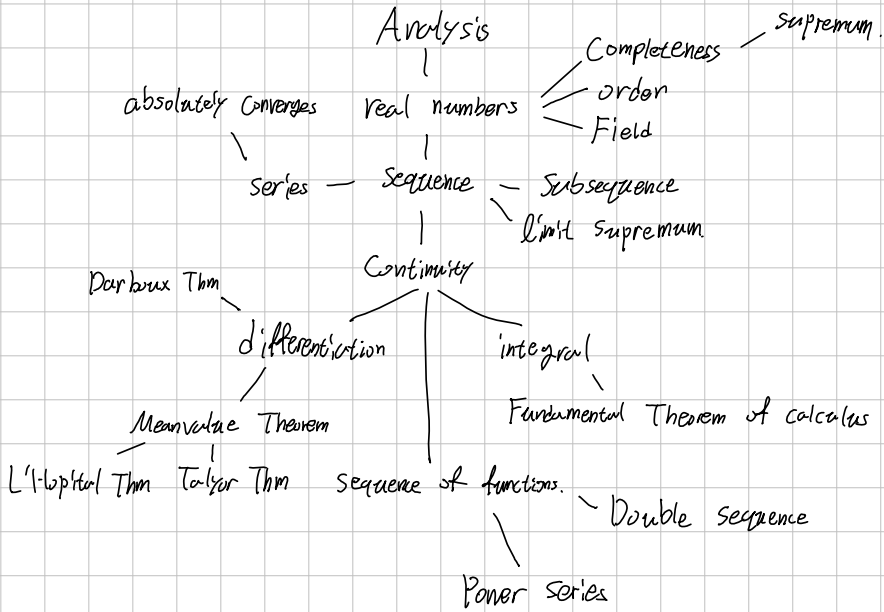
Solve, and obtimize in my own language.

Every study papers should be written under the fixed rules;
Prop: Proposition. A useful fact
Lemma: Lemma.
Thm: Theorem. A main result in the topic.
Cor.: Corollary.

Main Process to understand:

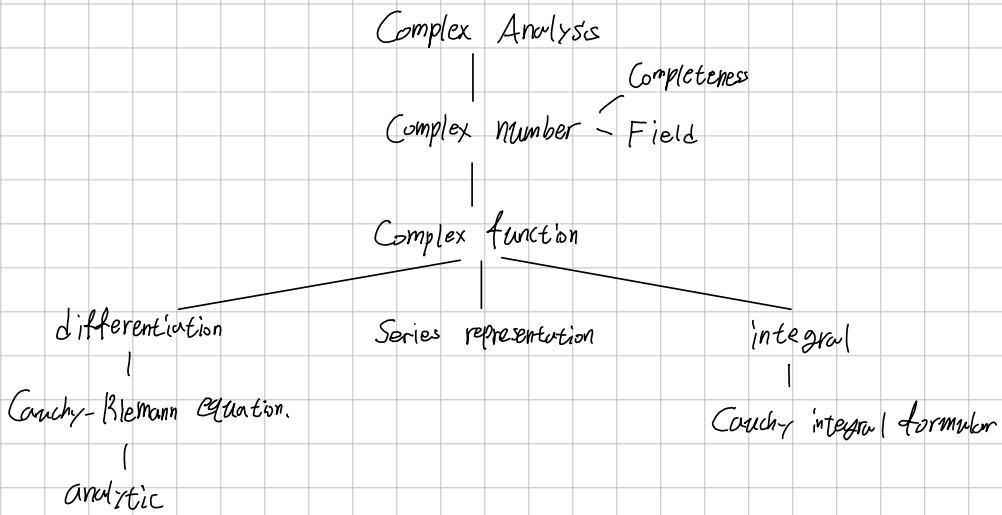

Linear Algebra.
Vector Space
basis
Span
linearly independent.
Existence
linear map
invertibility
determinant
triangularizable.
rational form
Jordan form
diagonalizable


Linear Algebra - View of Geometry
Inner Product.


Analysis
Completeness
Supremum
absolutely Converges
real numbers
order
Field
series
sequence
Subsequence
limit supremum
Continuity
Darboux Thm
differentiation
integral
Meanvalue Theorem
Fundamental Theorem of calculus
L'Hopital Thm
Taylor Thm
sequence of functions.
Double sequence
Power series

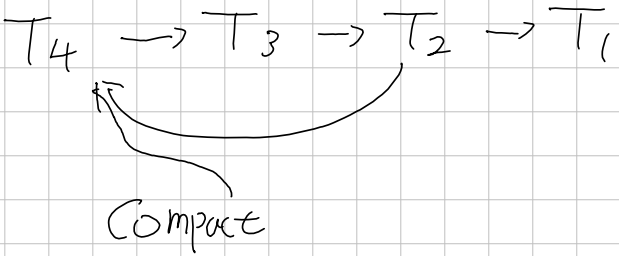

Algebra
Group
Ring
Field

Topology


Complex Analysis
Completeness
Complex number
Field
Complex function
differentiation
Series representation
integral
Cauchy-Riemann equation.
Cauchy integral formulas
analytic

Algebraic Topology

Differential Geometry


T_4
T_3
T_2
T_1
Compact